

Sketches: systematic view

U = universe of elements, $m = |U|$

randomly chosen $\text{RANK} : U \rightarrow [0..1]$ (real values)

s.t. for any $a, b \in U$, $a \neq b : \text{rank}(a) \neq \text{rank}(b)$

[if this condition is met, rank gives ^{an implicit} permutation of $U = \{u_1, u_2, \dots, u_m\}$
s.t. $\text{rank}(u_1) < \text{rank}(u_2) < \dots < \text{rank}(u_m)$]

Example: $\text{rank}(i) = \frac{i}{m}$ where $i \in [m]$ **NORMALIZED**

this takes considerably less space than storing explicitly
a random permutation

IN PRACTICE, using a random choice from a universal hash family is ok.

SKETCH | Set $A \subseteq U \mapsto$ its sketch $S(A)$ of A
[very often $S(A) \subseteq A$, so we assume this now]

Its supported operations:

- $\text{INIT}(A)$: build $S(A)$ from A [done only once at the beginning]
 as it requires scanning A
- UPDATE : $S(A \cup \{u\})$ obtained from $S(A)$, u instead of applying $\text{INIT}(A \cup \{u\})$
- UNION : $S(A \cup B)$ obtained from $S(A), S(B)$ (instead of INIT)
 expensive?
- SIZE : estimate of $|A|$ using $S(A)$
↳ (e.g. we can estimate $|A \cup B|$ using $S(A), S(B)$)

NOTE: sketch for $A \cap B$ is difficult

MIN-K SKETCH

Use k rank functions $r_1, r_2, \dots, r_k$

► INIT(A) : $S(A) = \{ \underbrace{\arg \min_{a_i \in A \text{ s.t. } r_i(a_i) \text{ is the min in } r_i(A)} r_i(A) : 1 \leq i \leq k \}$ (example we saw: second)

► UPDATE :
 $S(A \cup \{u\}) = \{ \arg \min \{ r_i(a_i), r_i(u) \}, 1 \leq i \leq k \}$

► UNION : $S(A \cup B) = \{ \arg \min \{ r_i(a_i), r_i(b_i) \}, 1 \leq i \leq k \}$

► SIZE : $\frac{k}{\sum_{1 \leq i \leq k} r_i(a_i)} - 1$ (i.e. we normalize each row)

$$S(A) = \begin{array}{cccc} r_1 & r_2 & \dots & r_k \\ \hline a_1 & a_2 & & a_k \end{array}$$

$$S(A) = \{a, c\}$$

$$S(A \cup \{a\}) = \{a, c, a\}$$

$k=3$

$a \mid a \mid b \mid c$
 $\uparrow$
 r_3

$a \mid c \mid b \mid a$
 $\uparrow$
 r_1

$b \mid a \mid c \mid a$
 $\uparrow$
 r_2

$c \mid a \mid b \mid a$
 $\uparrow$
 r_2

BOTTOM-K SKETCH

Just use a single rank function r

► INIT(A): $S(A) = \{\text{smallest } k \text{ elements according to } r(A)\}$
(if $|A| < k$, use a special element $\perp$ whose rank is $r(\perp) = 1$)

► UPDATE: $S(A \cup \{u\}) = \{\text{smallest } k \text{ elements in } S(A) \cup \{u\} \text{ according to } r(\cdot)\}$

► UNION: $S(A \cup B) = \{\text{smallest } k \text{ elements in } S(A) \cup S(B) \text{ according to } r(\cdot)\}$ $O(k)$ time

► SIZE: $\frac{k-1}{z_k}$ where $z_k = r(a)$ and a is the k -th element in $S(A)$

(intuitively: the larger $|A|$, the smaller z_k)

note that if $|A| < k$, we get $k-1$ as $z_k = r(\perp) = 1$)

however $S(A)$ has a meaning when $|A| > k$

THRESHOLD- t SKETCH

we a simple rank function, $t \in [0, 1]$

- ▶ INIT(A): $S(A) = \{a \in A : r(a) < t\}$ (observe that $|S(A)|$ can be large as it depends on t)
- ▶ UPDATE: $S(A \cup \{u\}) = \begin{cases} S(A) \cup \{u\} & \text{if } r(u) < t \\ S(A) & \text{o.w.} \end{cases}$
- ▶ UNION: $S(A \cup B) = S(A) \cup S(B)$
- ▶ SIZE: $\frac{|S(A)|}{t}$ ($t=1 \Rightarrow S(A)=A$)

As a running example, let's use JACCARD. Last lecture we saw a variant of MIN-k, so let's see here the other two.

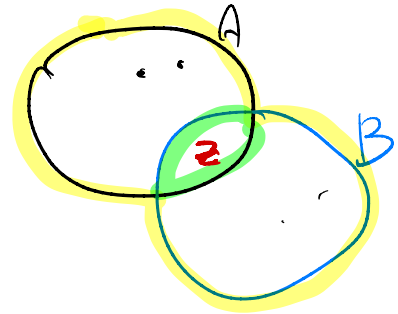
BOTTOM-k SKETCH for JACCARD INDEX $J(A,B) = \frac{|A \cap B|}{|A \cup B|}$

Given $X \subseteq U$, define $R_k(X) = \{\text{smallest } k \text{ elements in } X \text{ according to } r(\cdot)\}$

$$\text{Estimate } \tilde{J}(A,B) = \frac{|R_k(A \cup B) \cap (R_k(A) \cap R_k(B))|}{k}$$

$\overset{S(A \cup B)}{\#}$ $\overset{S(A)}{\underbrace{\hspace{1cm}}}$ $\overset{S(B)}{\underbrace{\hspace{1cm}}}$
 $R_k(S(A) \cup S(B))$ $R_k(A)$ $R_k(B)$

we need this set as an element $z \in R_k(A) \cap R_k(B)$ does not necessarily belongs to the first k elements in $A \cup B$, i.e. it is not necessarily true that $z \in R_k(A \cup B)$



example = $R_k(A) = \langle a, b, c, q \rangle$ $R_k(B) = \langle d, e, f, q \rangle$ $R_k(A) \cap R_k(B) = \{q\}$
 $k=1$ $R_k(A \cup B) = \langle a, b, c, d \rangle$

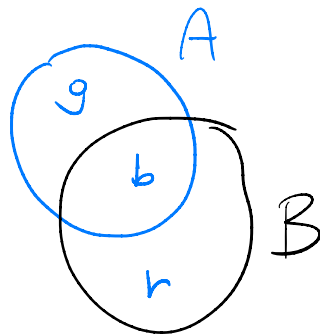
$r = \text{alphabetical order}$

We can rewrite $\tilde{J}(A,B) = \frac{|R_k(S(A) \cup S(B)) \cap S(A) \cap S(B)|}{k} \neq X$

We want to see that $\tilde{J}(A,B)$ is an unbiased estimator

$$E[\tilde{J}(A,B)] = J(A,B)$$

$$E\left[\frac{X}{k}\right] = J(A,B) \text{ is our goal}$$



obs Balls and bins situation:

- g elements in $A \setminus B$
- b elements in $A \cap B$
- r elements in $B \setminus A$

$$|A \cup B| = g + b + r$$

$$\Rightarrow J(A,B) = \frac{b}{g + b + r}$$

$$\Pr[X = c] = \frac{\binom{b}{c} \cdot \binom{r+g}{b-c}}{\binom{g+b+r}{b}}$$

$X = \#$ elements that belong to both $A \cap B$ and are in $R_k(S(A) \cup S(B))$

all possible choices of the remaining $b-c$ elements from the rest $(r+g)$ elements in $(A \cup B) \setminus (A \cap B)$

$$c \leq b$$

all possible choices of c elements out of b elements in the intersection

$$\Pr[X=c] = \frac{\binom{b}{c} \cdot \binom{r+g}{b-c}}{\binom{g+b+r}{b}}$$

since we do not know b ,
we just performed k times
the experiment (as it is Bottom- k)
we replace this b by k

Fortunately, this distribution is

known as hypergeometric and its expectation

$$E[X] = k \frac{b}{g+b+r} = k J(A, B) \Rightarrow \text{this motivates using } \frac{X}{k} \Rightarrow E\left[\frac{X}{k}\right] = J(A, B)$$

Example THRESHOLD- t with Jaccard index

$\tilde{J}(A, B) = \underbrace{J\left(\underbrace{S(A)}_{\text{available}}, \underbrace{S(B)}_{\text{sketches}}\right)}$ is an unbiased estimator

$$\begin{aligned} A' &= S(A) \\ B' &= S(B) \end{aligned} \quad \text{given } x \in A' \cup B' : Y_r^x = \begin{cases} 1 & \text{if } x \in A' \cap B' \\ 0 & \text{o.w.} \end{cases}$$

$$\text{We can rewrite } \tilde{J}(A, B) = \frac{\sum_{x \in A' \cup B'} Y_r^x}{|A' \cup B'|} = \frac{|A' \cap B'|}{|A' \cup B'|}$$

$$\Pr[Y_r^x = 1] \stackrel{\text{def}}{=} \Pr[x \in A' \cap B' \mid x \in A' \cup B'] =$$

$$= \underbrace{\Pr[x \in A' \cup B' \mid x \in A' \cup B']}_{=1 \text{ as } A' \cap B' \subseteq A' \cup B'} \cdot \frac{\Pr[x \in A' \cap B']}{\Pr[x \in A' \cup B']} = \frac{\Pr[x \in A \cap B] - \Pr[\cancel{rx} < t]}{\Pr[x \in A \cup B] \cdot \Pr[\cancel{rx} < t]} =$$

$$\Pr[E_1 | E_2] = \Pr[E_2 | E_1] \cdot \frac{\Pr[E_1]}{\Pr[E_2]} \quad (*)$$

A', B' were obtained by taking their elements with randomly assigned rank $< t$ (independently)

$$\frac{\Pr[x \in A \cap B]}{\Pr[x \in A \cup B]} \stackrel{(*)}{=} \Pr[x \in A \cap B | x \in A \cup B] = J(A, B)$$

Hence $\Pr[Y_r^x = 1] = J(A, B)$

As $\tilde{J}(A, B) = \frac{\sum_{x \in A' \cup B'} Y_r^x}{|A' \cup B'|}$, taking

$$E[\tilde{J}(A, B)] = \frac{\sum_{x \in A' \cup B'} E[Y_r^x]}{|A' \cup B'|} = \frac{\sum_{x \in A' \cup B'} J(A, B)}{|A' \cup B'|} = J(A, B)$$

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